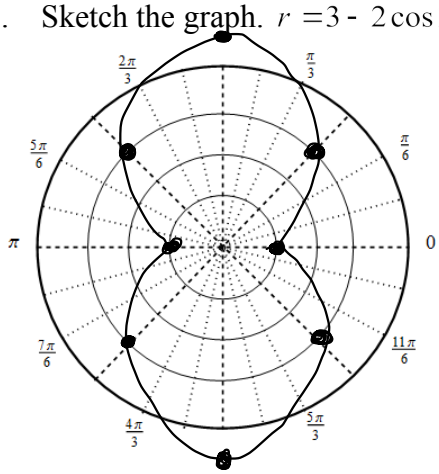
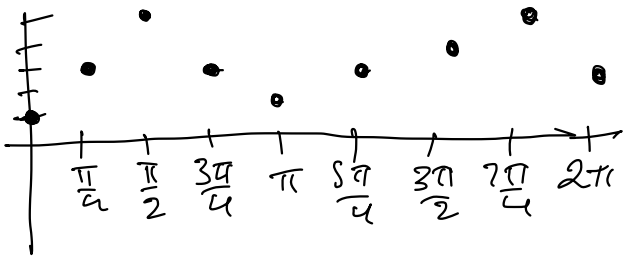
MATH 152 **QUIZ 11** NAME _____

Mrs. Bonny Tighe 25 points

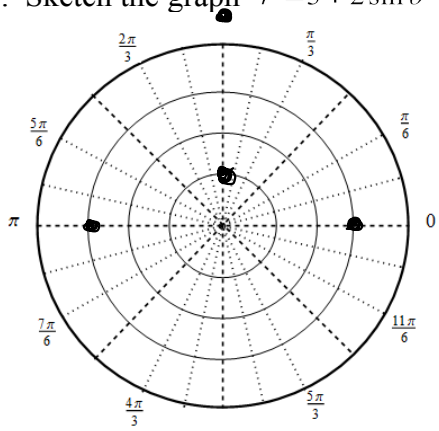
10.2, 10.3 TA_____ Wed 12/07/2022

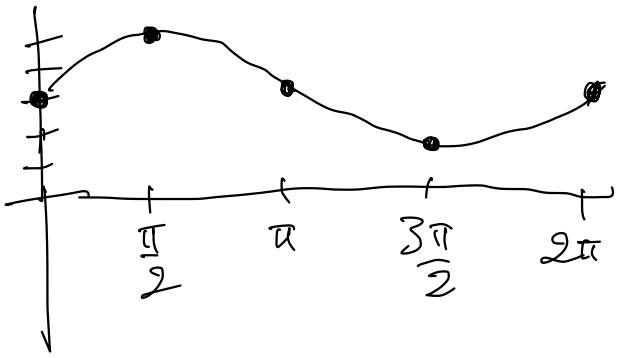
1. Sketch the graph. $r = 3 - 2\cos 2\theta$.



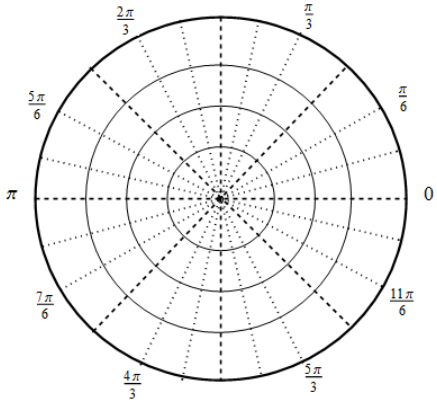


2. Sketch the graph $r = 3 + 2\sin\theta$





3. Sketch the graph $r = 2 + 3\sin\theta$



4. a) Find the Cartesian coordinates for the polar coordinate $(3, -7\pi/6)$

b) Find polar coordinates for the Cartesian coordinate $(-3\sqrt{3}, -1)$ where $r > 0, and\ \theta > 0$

c) Give three alternate versions for the polar point $(2, 5\pi/3)$

$r > 0,\ \theta < 0$ _____

$r < 0,\ \theta < 0$ _____

$r < 0,\ \theta > 0$ _____

5. How may petals would the graph of each of the following have?

a) $r = 3\sin 5\theta$ _____ b) $r = -4\cos 6\theta$ _____

6. The second derivative of the parametric equation $x = 2t + 1,\quad y = t^3 - 1$

A) $3t$

B) $3t/2$

C) $2/3$

D) $2t/3$

E) Some other answer

7) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\sin\theta, \quad y = 4\cos\theta$

b) (4 points) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

8) Compute the surface area obtained by rotating the given curve about the

y-axis $x = e^{t} + e^{-t}, \quad y = 5 - 2t, \quad 1 \le t \le 0$

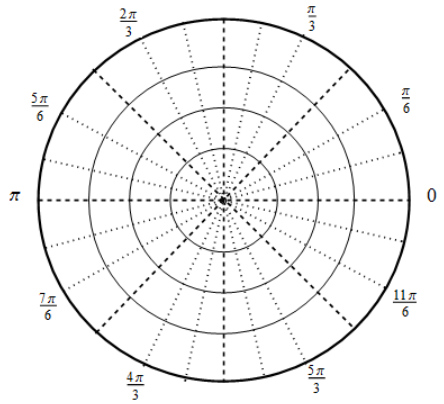
$$S = 2\pi \int_{\alpha}^{\beta} radius \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$$

MATH 152 **QUIZ 11A** NAME _____

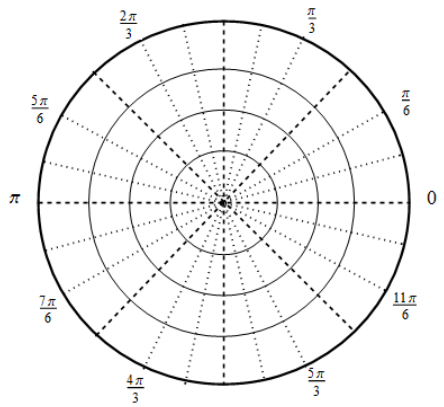
Mrs. Bonny Tighe 25 points

10.2, 10.3 TA_____ Wed 12/07/2022

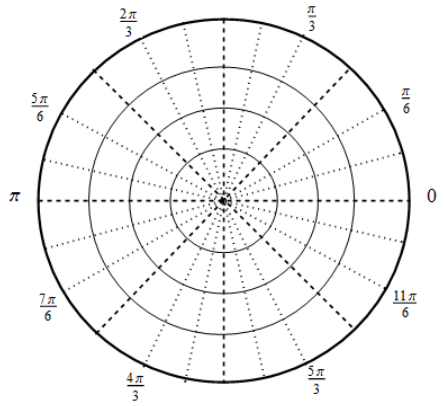
1. Sketch the graph. $r = 3 + 2\cos 2\theta$.



2. Sketch the graph $r = 2 - 3\sin\theta$



3. Sketch the graph $r = 2 + 2\cos\theta$



4. a) Find the Cartesian coordinates for the polar coordinate $(-2, -5\pi/6)$

b) Find polar coordinates for the Cartesian coordinate $(3, -3\sqrt{3})$ where $r < 0, and\ \theta > 0$

c) Give three alternate versions for the polar point $(4, 5\pi/4)$

$r > 0,\ \theta < 0$ _____

$r < 0,\ \theta < 0$ _____

$r < 0,\ \theta > 0$ _____

5. How may petals would the graph of each of the following have?

a) $r = 3\sin 4\theta$ _____ b) $r = -4\cos 7\theta$ _____

6. The second derivative of the parametric equation $x = 2t + 1,\quad y = t^3 - 1$

A) $3t$

B) $3t/2$

C) $2/3$

D) $2t/3$

E) Some other answer

7) a) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\sin\theta, \quad y = 4\cos\theta$

b) (4 points) Find the area under the curve on the interval $0 \le \theta \le \pi/2$

8) Compute the surface area obtained by rotating the given curve about the

y-axis $x = e^{t} + e^{-t}, \quad y = 5 - 2t, \quad 1 \le t \le 0$ $\qquad S = 2\pi \int_{\alpha}^{\beta} radius \sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$